



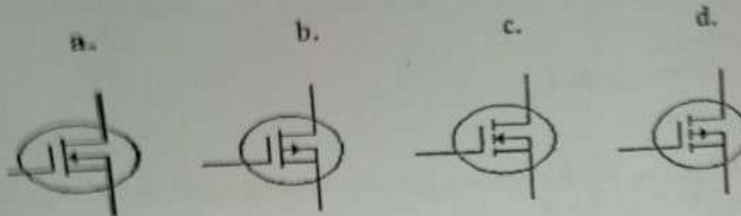
Student Name: أحمد يوسف طاهر
ابومندوب

Instructor Name:

1. Select the correct answer for the following questions:

The value of α when $\beta = 100$ is
a. 1.01 b. -101 c. 0.99 d. -9.9

2. Which figure represents a p-channel E-MOSFET?



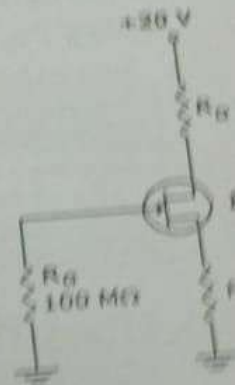
3. In BJT small signal model, the h_{21} parameter unit in hybrid equivalent model is.....
a. Ohms. b. Ampere. c. Volt. d. Unitless.

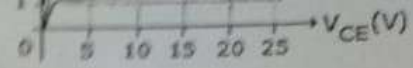
4. In BJT small signal model, the h_{11} parameter in hybrid equivalent model is calculated by
a. Making an open circuit at the input terminals.
b. Making an open circuit at the output terminals.
c. Making a short circuit at the output terminals.
d. Making a short circuit at the input terminals.

5. Using the data sheet for a BJT transistor, the parameter h_{FE} represents
a. α_{ac} b. β_{dc} c. β_{ac} d. α_{dc} e. None of them.

6. The best BJT configuration that provides the best Q- point stability with single-polarity supply is
a. Emitter follower. b. Collector-feedback. c. Voltage-divider. d. Fixed-bias.

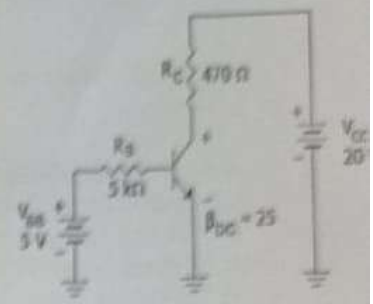
7. Using the figure shown below, the value of V_s is equal to.....
a. 6 V. b. 8 V. c. 20 V. d. 2 V e. None of them.





10. Which of the following is not correct about FET compared to BJT?
- a. FET is smaller size than BJT.
 - b. FET has a higher input impedance.
 - c. FET has Less temperature stability than BJT.
 - d. None of the above.

11. The collector-emitter voltage V_{CE} for the following configuration is equal to.....
- a. 19.3 V b. 9.2 V c. 0.7 V d. 9.9 V



12. In Shockley's equation for FET, the controlled variable that relates I_D and V_{GS} is.....
- a. I_{DSS} b. V_P c. I_D d. V_{GS}

13. For a FET transistor, the I_{DSS} for $V_{GS} = -4.5$ V, $V_P = -8$ V, and $I_D = 6.12$ mA is
- a. 32 mA b. 16 mA c. 64 mA d. 8 mA

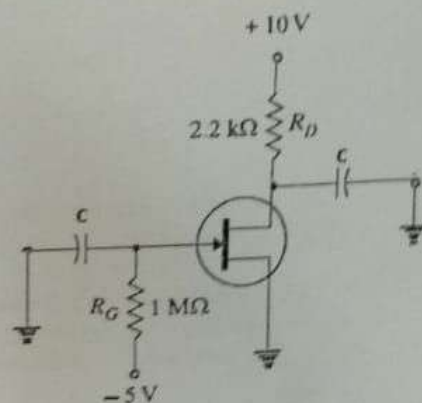
14. The data sheet for E-MOSFET is given below. For $V_{GS} (Th) = 3$ V, the parameter k of MOSFET is.....

ON CHARACTERISTICS				
Gate Threshold Voltage ($V_{GS} = 10$ V, $I_D = 10$ μ A)	$V_{GS(th)}$	1.0	5	V
Drain-Source On-Voltage ($I_D = 2.0$ mA, $V_{DS} = 10$ V)	$V_{DS(on)}$	-	1.0	V
On-State Drain Current ($V_{GS} = 10$ V, $V_{DS} = 10$ V)	$I_{D(on)}$	3.0	-	mA

- a. 6.17 mA/V^2 b. $6.17 \text{ V}^2/\text{mA}$ c. $0.061 \cdot 10^{-3} \text{ A/V}^2$ d. $0.061 \cdot 10^{-3} \text{ V}^2/\text{A}$

Q2. For the given circuit below, if $V_{GS(off)} = -8\text{ V}$, $I_{DSS} = 16\text{ mA}$, and $C = 2\text{ }\mu\text{F}$, answer the following:

1. What is the type of this transistor?
2. What is the name of this configuration?



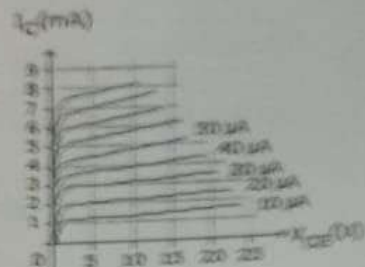
3. What is the value of V_{GS} ?
4. What is the value of I_D ?
5. What is the value of V_{DS} ?

Multiple Choice Answers of Question 1:

[illegible][illegible]

8. In JFET semi-conductor, when V_{as} is greater than V_p , the JFET has a characteristic of
- a. Voltage source. b. Current source. c. Ohmic region. d. None of them.

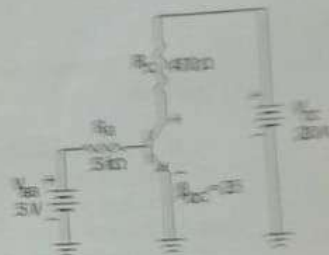
9. For the circuit shown below of collector characteristics, the value of β_{ac} for $V_{CE} = 15V$ and $I_B = 30 \mu A$ is equal to
- a. 160 b. 100 c. 50 d. 400



10. Which of the following is not correct about FET compared to BJT?

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- a. 32 mA b. 16 mA c. 64 mA d. 8 mA

14. The data sheet for E-MOSFET is given below. For $V_{GS(Th)} = 3V$, the parameter β_{eff} of MOSFET is

ON CHARACTERISTICS			
Gate Threshold Voltage ($V_{GS} = 10V, I_D = 10\mu A$)	$V_{GS(Th)}$	1.2	1.5
Drain-Source On-Voltage ($I_D = 2.0mA, V_{DS} = 10V$)	$V_{DS(on)}$	1.2	1.5
On-State Drain Current ($V_{GS} = 10V, V_{DS} = 10V$)	$I_{D(on)}$	1.2	1.5

- a. $6.17 mA/V^2$ b. $6.17 V^2/mA$ c. $0.061 \times 10^{-3} A/V^2$ d. $0.061 \times 10^{-3} V^2/A$

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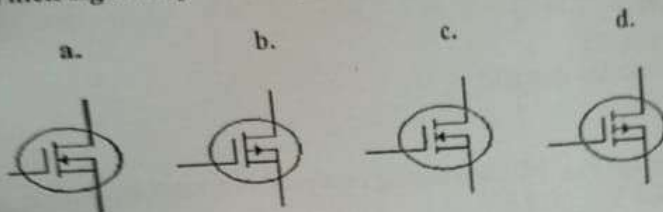
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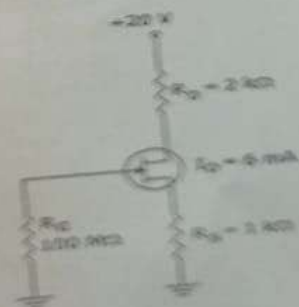
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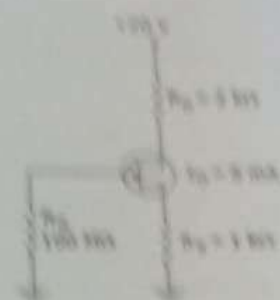
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a. 6 V. b. 8 V. c. 20 V. d. 2 V e. None of them.



14. For the following circuit, the value of V_{CE} is
 a. 21 V b. 2 V c. 2 V d. 21 V



15. For BJT transistor, the base-emitter junction in saturation region is
 a. Forward biased b. Reverse biased c. Ohmic region d. None of them

16. Assume that an electronic technician needs to design a circuit that makes the Q-point of the load line of the following circuit is going up toward the V_{CE} , then the value of should be
 a. R_E increased b. R_E increased c. R_E decreased d. R_E decreased e. R_E increased



17. The following BJT configuration is called configuration.
 a. Common base b. Emitter follower c. Emitter bias d. None of them



18. In a pnp transistor, the current carriers are
 a. Accepted ions b. donor ions c. electrons d. Holes

19. For the following BJT configuration, the base current is
 a. $I_B = \frac{V_{CE} - I_E R_E}{R_{B1} + (\beta + 1)R_E}$ b. $I_B = \frac{V_{CE} - I_E R_E}{R_E + (\beta + 1)R_E}$ c. $I_B = \frac{V_{CE} - I_E R_E}{R_{B1} + (\beta + 1)R_E}$ d. $I_B = \frac{V_{CE} - I_E R_E}{(\beta + 1)}$

